MAT41-4

Differential Equations

First- and second-order ODEs and their numerical solution

Continuous dynamics for finance and data science.

Albert School — 26 hours

Contents

1 ODE foundations

Definition 1 A **first-order ordinary differential equation (ODE)** in standard form is

$$y' = f(t, y),$$

where $y = y(t)$ is the unknown function and f is a given function of the independent variable t and the value y.

Definition 2 A **Cauchy problem** (initial value problem) couples an ODE with an initial condition:

$$y' = f(t, y), \quad y(t_0) = y_0.$$

Theorem 3 (Cauchy–Lipschitz) Let $f(t, y)$ be continuous on a rectangle around (t_0, y_0) and **Lipschitz in y**: there exists $L \geq 0$ such that

$$|f(t, y_1) - f(t, y_2)| \leq L|y_1 - y_2|$$

for all $(t, y_1), (t, y_2)$ in that rectangle. Then the Cauchy problem $y' = f(t, y)$, $y(t_0) = y_0$ has a unique solution on some interval around t_0.

A sufficient practical check: if $\partial f/\partial y$ exists and is continuous (bounded) near (t_0, y_0), then f is Lipschitz in y there, so existence and uniqueness hold.

> **Worked example.** For $y' = ty^2$ we have $\partial f/\partial y = 2ty$, continuous everywhere, so every Cauchy problem has a unique local solution.

1.1 Non-uniqueness when the condition fails

> **Worked example — y' = √|y|.** Take $y' = \sqrt{|y|}$, $y(0) = 0$. Here $f(y) = \sqrt{|y|}$ is continuous but **not** Lipschitz at $y = 0$ (its slope blows up there). Two solutions satisfy the same Cauchy problem:

$$y(t) = 0 \quad \text{and} \quad y(t) = t^2/4 \quad (t \geq 0).$$

One checks $\left((t^2/4)\right)' = t/2 = \sqrt{t^2/4}$. Uniqueness fails because the Lipschitz condition fails.

2 Separation of variables

A **separable** equation has the form $y' = g(t)h(y)$. When $h(y) \neq 0$, separate and integrate:

$$\int \frac{dy}{h(y)} = \int g(t)\,\mathrm{d}t.$$

Definition 4 An **equilibrium solution** is a constant $y(t) = c$ with $h(c) = 0$; then $y' = 0$ for all t.

Worked example. Solve $y' = ty$. Separating, $\int (dy)/y = \int t\,\mathrm{d}t$, so $\ln|y| = t^2/2 + C$, giving $y = Ke^{t^2/2}$. The equilibrium $y = 0$ (from $h(y) = y = 0$) is the case $K = 0$.

Common pitfall The formula $y = Ke^{t^2/2}$ is valid only on an interval where y keeps a constant sign, since dividing by $h(y)$ requires $h(y) \neq 0$. Always state the **domain of validity** of the solution.

3 First-order linear ODEs

A first-order **linear** ODE has the form

$$y' + a(t)y = b(t).$$

Proposition 5 (Integrating factor) Let $\mu(t) = e^{\int a(t)\,\mathrm{d}t}$. Multiplying the equation by μ makes the left side an exact derivative:

$$(\mu(t)y)' = \mu(t)b(t),$$

hence

$$y(t) = \frac{1}{\mu(t)}\left(\int \mu(t)b(t)\,\mathrm{d}t + C\right).$$

Worked example. Solve $y' + 2y = e^{-t}$. Here $a = 2$, $\mu = e^{2t}$, so $\left(e^{2t}y\right)' = e^{2t}e^{-t} = e^t$. Integrating, $e^{2t}y = e^t + C$, thus $y = e^{-t} + Ce^{-2t}$.

Remark **Variation of the constant.** Solve the homogeneous equation $y' + a(t)y = 0$ first, giving $y_h = Ce^{-\int a\,\mathrm{d}t}$. Then let the constant vary, $y = C(t)e^{-\int a\,\mathrm{d}t}$, substitute into the full equation, and solve for $C(t)$. This reproduces the integrating-factor result.

4 Second-order linear ODEs with constant coefficients

Consider

$$ay'' + by' + cy = g(t), \quad a \neq 0.$$

The general solution is $y = y_h + y_p$, where y_h solves the homogeneous equation ($g = 0$) and y_p is one particular solution.

4.1 Homogeneous case: the characteristic equation

Seeking $y = e^{rt}$ in $ay'' + by' + cy = 0$ gives the **characteristic equation**

$$ar^2 + br + c = 0, \quad \text{discriminant} \;\; \Delta = b^2 - 4ac.$$

Proposition 6

- **$\Delta > 0$ (distinct real roots r_1, r_2):** $y_h = C_1 e^{r_1 t} + C_2 e^{r_2 t}$.
- **$\Delta = 0$ (repeated root r):** $y_h = (C_1 + C_2 t)e^{rt}$.
- **$\Delta < 0$ (complex roots $r = \alpha \pm i\beta$):** $y_h = e^{\alpha t}(C_1 \cos(\beta t) + C_2 \sin(\beta t))$.

Worked example.

- $y'' - 3y' + 2y = 0$: roots $1, 2$, so $y = C_1 e^t + C_2 e^{2t}$.
- $y'' - 2y' + y = 0$: repeated root 1, so $y = (C_1 + C_2 t)e^t$.
- $y'' + 4y = 0$: roots $\pm 2i$, so $y = C_1 \cos(2t) + C_2 \sin(2t)$.

4.2 Non-homogeneous case: undetermined coefficients

Guess a particular solution y_p of the same form as $g(t)$, with unknown coefficients fixed by substitution.

$g(t)$	**Trial** y_p
$P(t)$ (polynomial of degree n)	polynomial of degree n
e^{kt}	Ae^{kt}
$\cos(\omega t)$ or $\sin(\omega t)$	$A\cos(\omega t) + B\sin(\omega t)$

Common pitfall If the trial form already solves the homogeneous equation, multiply it by t (or t^2 for a repeated root) before fixing the coefficients.

Worked example. Solve $y'' - 3y' + 2y = e^{3t}$. Try $y_p = Ae^{3t}$: $(9 - 9 + 2)Ae^{3t} = e^{3t}$, so $A = 1/2$. With the homogeneous part above, $y = C_1 e^t + C_2 e^{2t} + \frac{1}{2}e^{3t}$.

5 Numerical methods

When no closed form is available, approximate the solution of $y' = f(t, y)$, $y(t_0) = y_0$ on a grid $t_n = t_0 + nh$ with step h.

5.1 Euler's explicit method

$$y_{n+1} = y_n + hf(t_n, y_n).$$

Worked example — By hand. For $y' = y$, $y(0) = 1$, $h = 0.5$: $y_1 = 1 + 0.5 \cdot 1 = 1.5$, $y_2 = 1.5 + 0.5 \cdot 1.5 = 2.25$. The exact value is $e^1 \approx 2.718$, so y_2 underestimates it.

> **Definition 7** The **local truncation error** is the error introduced in one step assuming exact input; the **global error** is the accumulated error at a fixed final time T. A method has **convergence order p** if the global error is $O(h^p)$.

Euler's method has order $p = 1$: halving h roughly halves the global error. This can be checked empirically by computing the error at the final time for $h, h/2, h/4, \ldots$ and verifying it falls by a factor ≈ 2 each time.

5.2 Runge–Kutta (RK4)

$$\begin{aligned} k_1 &= f(t_n, y_n), \\ k_2 &= f(t_n + h/2, y_n + hk_1/2), \\ k_3 &= f(t_n + h/2, y_n + hk_2/2), \\ k_4 &= f(t_n + h, y_n + hk_3), \\ y_{n+1} &= y_n + \frac{h}{6}(k_1 + 2k_2 + 2k_3 + k_4). \end{aligned}$$

RK4 has convergence order $p = 4$: halving h divides the global error by about 16. For the same accuracy it needs far fewer steps than Euler, at the cost of four evaluations of f per step. The step size h is chosen as a trade-off between accuracy (smaller h) and cost (larger h).

6 Python implementation

The same problems are solved in Python. With **NumPy** one stores the grid and iterates the Euler or RK4 update in a loop over `numpy` arrays.

SciPy provides `solve_ivp`, which integrates a system numerically:

```python
import numpy as np
from scipy.integrate import solve_ivp

def f(t, y):
    return y          # y' = y

sol = solve_ivp(f, (0, 1), [1.0], method="RK45",
                t_eval=np.linspace(0, 1, 11))
# sol.t, sol.y hold the grid and the solution
```

`solve_ivp` accepts a vector-valued `f` for a **system** of ODEs, supports several methods (`"RK45"`, `"RK23"`, `"Radau"`, …), and returns trajectories that can be plotted and compared across methods.

7 Finance and growth applications

> **Worked example — Continuous compound interest.** Capital at rate r: $y' = ry$, $y(0) = y_0$. Separable/linear, with solution $y(t) = y_0 e^{rt}$.

> **Worked example — Portfolio with contributions.** Adding a contribution rate $c(t)$: $y' = ry + c(t)$. This is first-order linear; solve with the integrating factor $\mu = e^{-rt}$.

> **Worked example — Logistic growth.** $y' = ry(1 - y/K)$ is separable with equilibria $y = 0$ and $y = K$. Its solution is

> $$y(t) = \frac{K}{1 + ((K - y_0)/y_0)e^{-rt}}.$$

> **Worked example — Debt repayment.** Balance under interest r and repayment rate p: $y' = ry - p$, a first-order linear ODE solved by the integrating factor.

Each model is solved analytically and then validated numerically (Euler/RK4 or `solve_ivp`), confirming the closed form.

8 Method selection

Analytical vs. numerical. Use an analytical solution when the equation is separable, first-order linear, or second-order linear with constant coefficients — these have known closed forms. When f is nonlinear or has no tractable antiderivative, an analytical solution is unrealistic and a numerical method (Euler for simplicity, RK4 or `solve_ivp` for accuracy) is the appropriate choice. Always assess tractability before committing to an approach.